# ProteinNetworkSlight Tutorial

Tutorial PDF

## Summary

ProteinNetworkSlight is a web-based tool for constructing patient-specific protein interaction networks from user-provided gene or protein scores. The platform integrates quantitative input data with interaction information retrieved from the STRING database to identify biologically relevant connections. The resulting networks are further linked to the Cancer Drugs Database, enabling the exploration of potential therapeutic targets.

Tutorial Video

ProteinNetworkSlight integrates information from two external databases:

- STRING v9 - a database of known and predicted protein-protein interactions.
- Cancer Drugs Database - a resource containing licensed anticancer drugs.
- UniProt - nucleotide sequence and functional information resource.

The generated networks allow interactive exploration of protein interactions and facilitate the identification of biologically important proteins and potential drug targets.

## Workflow Overview

The ProteinNetworkSlight workflow consists of three main steps:

1. Upload input data - Users upload a tabular file containing protein or gene identifiers together with numeric scores (e.g., DE or DAS values, p-values, or other quantitative measures).
2. Network construction - The server maps the identifiers to the identifiers contained in STRING, retrieves interaction data, and organizes it into a network.
3. Explore the resulting network - The platform constructs a network interactome using interaction data from the STRING database. Users can interactively explore the network, adjust visualization settings, and search the results for genes of interest or disease formats.

## Starting Point

To begin the analysis, users must upload an Excel, CSV or TSV containing a single sheet of tabular data. Each row should correspond to a protein or gene, and each column should represent a feature. The first column must contain unique identifiers, followed by one or more numeric columns containing quantitative values. Additional columns should contain textual or other non-numeric data such as gene descriptions or annotations.

The first row must contain column headers. Numeric columns intended for analysis must share a common prefix.

An example file is provided (adapted from Vasudevan et al., ag Precision Oncology, 2021, Supplementary Data 5). In this example, the gene identifier column is labeled "GN", and the numeric columns begin with the prefix "DE". Followed by an index.

In this dataset, the 3-dimensional transport protein weights derived from the information theoretic analysis described in Vasudevan et al. However, the format is flexible, and these columns may contain any quantitative scores, coefficients, or fold changes values, depending on the biological question identify or expression patterns.

| GN | DE_1 | DE_2 | DE_3 | DE_4 | DE_5 |
|---|---|---|---|---|---|
| A1BG | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| A1CF | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| A2M | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| A2ML1 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| A3GALT2 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| A4GALT | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| AAAS | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| AAGAB | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| AAK1 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| AAMP | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| AANAT | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| AARS | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| AARS2 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| AARSD1 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |

## Setting parameters and thresholds

In this step the user should specify:

- Identifier column - The column with unique protein/gene identifier (column 0 by default).
- Numeric columns - Columns with numeric values (e.g., DE, DAS, or other scores) (e.g., "DE" where DE represents the weight of each protein/gene in the network (e.g. DE, DAS).
- STRING interaction threshold - The lower bound for the STRING interaction score, representing the probability of a specified interaction.
- Score thresholds - Positive and negative thresholds used to determine whether scores fall within a specified range.
- Organism of interest - For example, Homo sapiens in the example dataset.

## Manual thresholds adjustments

- You can also choose to adjust the thresholds of every score (DE in the example) column in the graph individually.

## Adjusting prototypiqne names

ProteinNetworkSlight retrieves interaction data from STRING v9. In order to construct and visualize protein interaction networks. Therefore, protein or gene identifiers in the input file should match the names used in STRING. Mismatches may result from multiple synonyms, some identifiers in the input file may not match automatically. The following steps highlight such cases and allow users to select alternative identifiers.

- Upon uploading a file, the system attempts to match identifiers to STRING entries. If some identifiers fail to match, a pop-up window will appear, displaying the number of unmatched identifiers and suggesting alternatives.

Protein Input
Protein Input
Protein Input